Yihao Xu

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Research Profile

- Computational design science researcher working on multimodal learning and predictive analytics for digital platforms, with additional work in computational measurement of organizational culture.
- First author of a paper at the *International Conference on Information Systems* (2025) and an article in *IEEE Sensors Journal*; the ICIS paper was a Best Complete Paper Nominee / Best-in-Track Paper.
- Experience designing multimodal and multi-view learning models, modeling temporal distribution shift, and measuring constructs with neural text embeddings; Python, PyTorch, and SQL.
- Teaching assistant across two fall terms; coordinated six study sessions involving 60 participants who are deaf or hard of hearing in an AR accessibility study.

Education

M.S. in Management Science and Engineering **Expected 2027**
School of Management, University of Science and Technology of China (USTC) *Hefei, China*
Advisor: Prof. Hefu Liu.

B.M. in Information Management and Information Systems **2024**
School of Management, Hefei University of Technology (HFUT) *Hefei, China*

Research Interests

Topics: Digital platforms and crowdfunding · Cultural and creative markets · Organizational culture · Accessible human–AI interaction
Methods: Multimodal representation learning · Multi-view deep clustering · Temporal generalization · Computational text measurement

Published Research

1. Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction
Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao.
46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, 2025.
ICIS 2025 Best Complete Paper Nominee / Best-in-Track Paper.

- Examines how cultural values embedded in project content can be represented and combined for pre-launch crowdfunding success prediction.
- CV-PM turns Throsby's six cultural values into features from metadata, text, and images. Convolutional layers refine the features, and attention combines the value representations for prediction.
- Tests on 14,539 Kickstarter projects yielded AUCs of 94.72% in Art and 95.14% in Design, exceeding the strongest baseline in each category by 1.49–3.65 percentage points.

2. Multisensor Deep Clustering Networks With Joint Learning Strategy for Fault Diagnosis Considering Consensus-Specific Constraint
Yihao Xu, Mingfeng Lu, Gang Wang, and Fang Fang.
IEEE Sensors Journal, 25(11), 19921–19936, 2025. doi:10.1109/JSEN.2025.3560460.

- In two-stage unsupervised fault diagnosis, extracted features are not optimized for clustering. This work learns across sensor views and trains feature extraction and clustering end to end.
- MsDCN uses a sensor-attention encoder for shared patterns and separate encoders for sensor-specific signals. Consensus-specific constraints retain both types of information during joint training.
- Achieved the highest mean clustering accuracy among the tested methods on six gearbox and bearing datasets, with gains of 19.83–28.41 percentage points over K-means. Ablations supported the sensor-attention and consensus-specific designs.

Work in Progress

3. Multimodal Learning under Evolving Environments for Crowdfunding Outcome Prediction
Yihao Xu, Wenxiang Zhao, and Hefu Liu. *Planned submission to Information Systems Research in 2026.*

- Pre-launch prediction must combine shared and modality-specific campaign information while allowing its relationship with fundraising outcomes to change over time.
- The model organizes text and image information by semantic dimension. A temporal component updates each dimension's predictive contribution using historical environments and target-campaign information.
- Evaluated on 54,306 Kickstarter campaigns in four categories, using a fixed chronological split and 20 training seeds per category. With validation-based model selection, AUROC improved by 3.55–5.63 percentage points over the strongest of nine benchmarks in each category.

4. From Anchor to Evidence: A Theory-Driven Model for Newly Launched Crowdfunding Success Prediction
Wenxiang Zhao, **Yihao Xu**, and Xiaoyu Yao. *Working Paper.*

- Crowdfunding platforms show brief project summaries before detailed pages. Pooling these inputs overlooks the distinction between an initial impression and later evidence.
- The model follows anchoring-and-adjustment theory: summary cues form an anchor, then project details and comparisons with similar projects provide evidence for adjustment.
- The model reached AUCs of 94.57% on 13,192 Technology projects and 83.30% on 19,744 Art projects, outperforming all reported baselines.

5. Scale-Anchored Neural Measurement of Firms' Competing Values
Research in progress; measurement stage completed.

- Culture surveys offer limited coverage across firms and years, while annual reports use language that differs from survey items. The project adapts OCAI-based measurement to firm-year text data.
- Semantic retrieval and expert review select report excerpts for each scale item. Scoring retains the item structure rather than pooling all excerpts into a single representation.
- The protocol can be reused with other constructs and encoders. It has produced firm-year scores for four Competing Values Framework orientations from 4.37 million MD&A segments spanning 2001–2025.

6. AI Assistants and Customer Satisfaction in Digital Service Platforms
Early-stage Research Project.

- Proposed study of how conversational behavior in Meituan customer–AI assistant dialogues relates to user satisfaction, using dialogue records and satisfaction outcomes.

Conference Presentation

Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao. *Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction.* 46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, Dec. 2025.

Research and Teaching Experience

Research Assistant, University of Science and Technology of China *Jun. 2026–Present*
Supported an AR smart-glasses study on communication access and social inclusion for people who are deaf or hard of hearing; coordinated materials, participant arrangements, device setup, and six study sessions involving 60 participants.

Teaching Assistant, University of Science and Technology of China *Fall 2025, 2026*
Information Technology and Organizational Strategy 42 students; 40 hours
Instructor: Prof. Juan Qin; coordinated assignments, group activities, and feedback.

Awards and Scholarships

Awards

- Outstanding Graduate of Anhui Province (top 3% students) 2024
- Outstanding Student Award (top 10% students) 2023

Scholarships

- China National Scholarship 2025
- First-Class Graduate Scholarship, USTC 2025, 2026
- Shishengqing Scholarship, School of Management, HFUT 2024
- Academic Progress Scholarship, School of Management, HFUT 2022
- First-Class Scholarship, HFUT (top 3% students) 2021

Internships

Research Intern *Jun.–Aug. 2026*
USTC Yangtze River Delta Science and Technology Strategic Frontier Research Center
Conducted desk research, case analysis, and comparative studies of firms' early development and commercialization in commercial spaceflight, AI, and semiconductors.

Technical Service Intern, CEO Office, Volkswagen (Anhui) Automotive Co., Ltd. *Apr.–Jul. 2024*
Hefei, China. Built and maintained an automotive after-sales database for recurring service reporting and operational analysis.

Computational Skills and Languages

Computational Methods: Multimodal learning · Multi-view deep clustering · Attention mechanisms · Temporal generalization · Sentence embeddings and semantic retrieval
Programming: Python · PyTorch · SQL · Stata
Languages: Chinese (native) · English (IELTS Academic 7.5)